Natalie Truesdell

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SUMMARY

Data Science graduate student at the University of British Columbia with experience developing machine learning models, building scalable data pipelines, and performing statistical analysis on large-scale datasets. Skilled in Python, SQL, data visualization, feature engineering, and predictive modeling. Experienced communicating technical findings to diverse stakeholders through reports, dashboards, and presentations. Passionate about applying data-driven solutions to real-world engineering and operational challenges.

SKILLS

Languages & Databases: Python, SQL, R, Bash, Git

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, PyTorch, feature engineering, supervised & unsupervised learning, deep learning, model evaluation, cross-validation, hyperparameter tuning

Data Engineering & Cloud: GCP (Vertex AI, GCS, BigQuery), AWS, DuckDB, PyArrow, Docker, CI/CD, GitHub Actions, data pipelines, workflow automation, data validation, QA/testing

Statistics: Hypothesis testing, regression modeling, Bayesian inference, experimental design, causal inference, A/B testing, time series analysis

Data Visualization & Reporting: Matplotlib, Seaborn, Tableau, Shiny, PowerPoint, Excel

EDUCATION

University of British Columbia, Vancouver | *Master of Data Science* *Aug 2025 – Jun 2026*

Cumulative Average: 92%

University of California, San Diego | *B.S. Business Economics, Minor in Data Analytics* *Sept 2020 – Jun 2025*

GPA: 3.6/4.0

EXPERIENCE

Data Analytics Intern | *Acorn Evaluation, Inc.* Jan 2025 – Jul 2025

- Extracted, cleaned, and transformed multi-source datasets (30K+ records) using reproducible Python workflows
- Developed a Python-based web scraping and parsing pipeline to transform unstructured data from 200+ program websites into an analysis-ready dataset
- Maintained and validated interactive Tableau dashboards for 5+ clients to monitor program metrics and support operational decision-making
- Implemented anomaly detection and data QA processes, identifying inconsistencies across datasets and improving standardization and reusability across reporting workflows

Accounting Intern | *Lohman & Associates* Jun 2024 – Nov 2024

- Analyzed financial and operational datasets to produce weekly Excel reports on revenue forecasts and team KPIs
- Identified and resolved data discrepancies, improving accuracy, consistency, and standardization of reporting workflows

PROJECTS

Smart Amazon Product Query | *LLM-powered Retrieval and Evaluation System* GitHub

- Built a scalable retrieval system over 2.5M+ records using hybrid search (BM25 + FAISS embeddings)
- Developed an LLM-powered RAG pipeline with prompt engineering, grounding guardrails, and source attribution to improve recommendation reliability and reduce hallucinations
- Optimized large-scale data querying using DuckDB and Parquet-based processing for low-latency analytics workflows

Multimodal Patient Outcome Prediction | *UBC MDS Capstone · Predictive modeling pipeline*

- Designed an end-to-end predictive modeling pipeline integrating 3D CT imaging with high-volume EHR data (225M+ rows) to predict patient mortality and hospital readmission
- Investigated model outputs using feature importance and explainability methods, including Grad-CAM heatmaps and slice-level gradient attribution to identify key clinical regions and quantify their contribution to predictions
- Streamlined data ingestion using DuckDB and PyArrow for efficient GCP bucket querying and managed the full model lifecycle and metadata tracking via Vertex AI Experiment Tracking
- Presented modeling results and performance tradeoffs to clinical stakeholders, translating technical findings into actionable recommendations

NYC Squirrel Behavior Dashboard | *Interactive analytics dashboard for behavioral insights* Dashboard • GitHub

- Conducted exploratory analysis on behavioral and geographic patterns within a large public dataset
- Built an interactive dashboard enabling segmentation, filtering, and trend analysis across multiple dimensions
- Implemented natural-language querying functionality to support self-service analytics and insight generation

SurveyCleaner Python Package | *Production ML data-cleaning library with full MLOps pipeline* TestPyPI • Github

- Designed and implemented a Python package for automated data cleaning, validation, and preprocessing workflows
- Built a comprehensive Pytest framework achieving 100% test coverage across validation scenarios and edge cases
- Developed CI/CD workflows using GitHub Actions for automated testing, versioning, and deployment to TestPyPI